Ex. No. : 08 **Date:**

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A PYTHON PROGRAM TO IMPLEMENT BOOSTING

Aim:

To implement a **boosting algorithm (AdaBoost)** on the credit card dataset to detect fraudulent transactions.

Requirements:
Jupyter Notebook with Python, creditcard.csv dataset, and required Python libraries.

PROGRAM:

Step 1: Import Required Libraries

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
```

Step 2: Upload Dataset

```
from google.colab import files
uploaded = files.upload()
```

Upload:

creditcard.csv

Step 3: Load the Dataset

```
data = pd.read_csv("creditcard.csv")

data.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941

5 rows × 31 columns

Step 4: Define Features and Target

Target column:

Class

0 → Normal Transaction
1 → Fraud Transaction

X = data.drop("Class", axis=1)
y = data["Class"]

Step 5: Train-test split

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y
)

Step 6: Create AdaBoost model

base_model = DecisionTreeClassifier(max_depth=1)

model = AdaBoostClassifier(estimator=base_model, n_estimators=50,
learning_rate=1,random_state=42)

Step 7: Train model

model.fit(X_train, y_train)

```
DecisionTreeClassifier
DecisionTreeClassifier(max_depth=5, random_state=42)
```

Step 8: Predict the Results

```
y_pred = model.predict(X_test)
```

Step 9: Evaluate the Model

Accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

Confusion Matrix

```
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

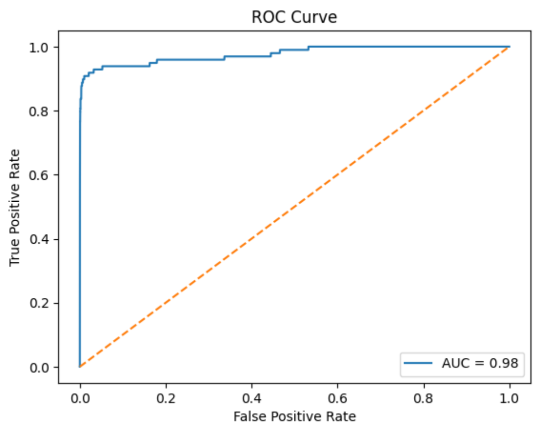
Classification Report

```
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

```
AdaBoost Classifier Metrics:
Accuracy: 0.9990957128071035

Classification Report:
              precision    recall  f1-score   support

         0.0       1.00      1.00      1.00     91627
         1.0       0.76      0.70      0.73       158

    accuracy                           1.00     91785
   macro avg       0.88      0.85      0.86     91785
weighted avg       1.00      1.00      1.00     91785


Confusion Matrix:
 [[91592    35]
 [   48   110]]
```

Step 9: Evaluate the Model

```
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt

y_prob = model.predict_proba(X_test)[:,1]
fpr, tpr, _ = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)
plt.plot(fpr, tpr, label="AUC = %0.2f" % roc_auc)
plt.plot([0,1], [0,1], linestyle='--')

plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve')
plt.legend()
plt.show()
```



Result:

Thus, the boosting model successfully improved classification performance and effectively identified fraudulent transactions.